

Teaching Statement — Chris Smith

Course-based Undergraduate Research Experiences (CUREs)

CUREs teach students course material while engaging them in real research. Using real data to make genuine discoveries in the classroom increases student engagement, biology major retention, recruitment into research labs, and may simultaneously advance the instructor's research program. Importantly, CUREs make research accessible to students who work for tuition or face other challenges with accessing research opportunities. I had the opportunity to design and teach a CURE, which was an independent lab section associated with “Introduction to Research in Ecology and Evolution”. In my lab students identified local apple tree cultivars using molecular markers and analyzed genetic diversity, population structure, selection for cold-hardy varieties, and the genetic basis of fruit coloration. I also received formal training in CUREs from a CURE-design class offered by Dr. Lisa Corwin at CU. In this class, I studied CURE examples and amassed a toolbox of evidence-based teaching strategies that I used in my own course.

Data proficiency

There is a flood of biological data currently being produced—e.g., nucleotide sequencing, image data—and it is critical for the next generation of scientists to fully leverage these unwieldy, but information-rich, datasets. My quantitative and computational specialization, and my experience as a teaching assistant for Biological Statistics, let me equip trainees with such skills. Specifically, I will train graduate students to program in Python and R, apply statistics, work in high performance computing environments, contribute to open-source software, and build machine learning models.

Specialized course option: Neural network applications in biology

I would be very excited to teach a neural network applications in biology course. My vision is to teach a course where students (1) learn how neural networks work, (2) learn to implement several kinds of neural networks for analyzing biological data, and (3) develop a machine learning repertoire—knowing which tools are available and which to reach for in each situation. Each week we will build models with gradually increasing complexity: neural network machinery will be introduced one piece at a time. For example, Week-1 students will hand-code an ordinary least squares (OLS) regression from scratch; in the next assignment, we will swap out the simple formula used for solving the OLS problem, and swap in a loss function and optimization algorithm—these are some basic components of neural networks that can solve the same problem in a more automated way. In subsequent weeks will add on additional pieces until have have a complete neural network, and in the later weeks we will cover a few different kinds of neural networks. Throughout the semester, we will (i) compare and contrast each new technique with previously covered models; (ii) analyze biological data such as health data, wildlife images, genomic data, and amino acid sequences; (iii) cover enough math for students to gain an intuition for how machine learning algorithms work while remaining accessible to biology majors; (iv) work on a class project, where students

apply machine learning to their own or publicly available data; and (v) participate in computer lab assignments and group work.

Making undergraduate research more accessible

Serving on the DEI Committee in the Biology Department at UO, I was part of a sub-committee for making undergraduate research more inclusive. For this aim, I produced a tri-fold brochure advertising undergraduate research. My intension with the brochure was not simply recruitment, but to promote diversity by recruiting students from groups that are underrepresented in biology research. For example, I wanted to show first-generation college students that exciting research opportunities are actually available. Likewise, it is important to inform students who work for their tuition that paid research positions exist, and that working in a lab can earn course credit. At Indiana University I would look for similar avenues for making undergraduate research more inclusive.